CUTUBKA 2AAD · CASHARKA 3AAD EE 7-DA CASHAR

Data Science

DHAMMAADKA CASHARKAN

Waxaad awoodi doontaa inaad sharaxdo waxa Data Science qabto, natiijooyinka uu soo saari karo, iyo sababta aanu mar walba ugu baahnayn Machine Learning.

Dulmar

Shirkad isgaarsiineed ayaa aragtay in kumannaan macaamiil ahi joojiyeen cusboonaysiinta xirmooyinkooda billaha ah. Diiwaannadu waxay muujinayaan waxa dhacay, laakiin shirkaddu waxay u baahan tahay inay fahanto sida dhibaatadu u dhacday, tijaabiso sababaha suurtagalka ah, oo go'aamiso caddaynta taageeri karta xal.

Data Science wuxuu waxtar u leeyahay dhibaato noocan ah oo aan jawaabteedu markiiba caddayn.

Waa maxay Data Science?

Data Science wuxuu isku daraa tirakoob (statistics), barnaamij-samaynta (programming), aqoonta goobta (domain knowledge), iyo falanqaynta xogta si xogta looga helo faham muhiim ah loona dhiso xalal xog ku salaysan.

Qayb walba waxay keentaa wax gaar ah:

QAYBTA	WAXA AY KU KORDHISO
Tirakoob (statistics)	Habab lagu cabbiro qaababka soo noqnoqda (patterns), hubanti la'aanta (uncertainty), iyo caddaynta
Barnaamij-samaynta (programming)	Hab xog badan ama adag loogu shaqayn karo si la isku halayn karo
Aqoonta goobta (domain knowledge)	Fahamka dhibaatada dhabta ah iyo xaddidaadaha jira
Falanqaynta xogta	Habab xogta lagu baaro laguna sharxo waxa ay muujinayso

Fahamka ganacsiga iyo fahamka tirooyinku si isku mid ah ayay muhiim u yihiin. Qaab (pattern) ka dhex muuqda mobile data macne badan ma leh haddii qofka baaraya aanu fahmin xirmooyinka, qiimahooda, ama hab-dhaqanka macaamiisha.

Waxa uu Qabto Data Scientist-ku

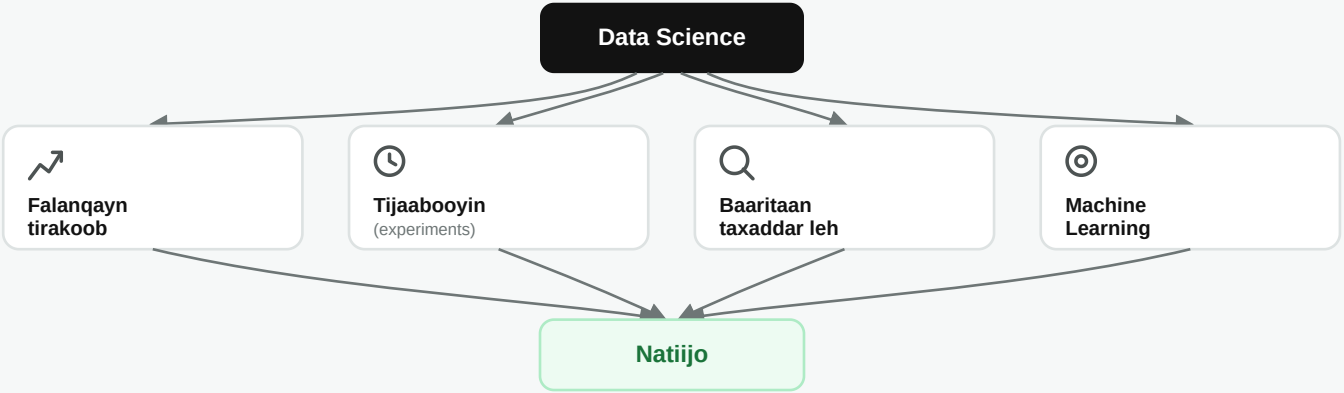
Data Scientist-yadu waxay badanaa ka shaqeeyaan dhibaatooyin ay ka mid yihiin:

- Saadaalinta baahida alaabta
- Ogaanshaha khiyaanada (fraud detection)
- Diyaarinta iyo qiimaynta tijaabooyinka (experiments)
- Dhisidda alaabo xog ku shaqeeya (data products) ama model saadaalin sameeya (predictive model)

Shaqadu badanaa waxay ka bilaabataa su'aal, kama bilaabato model. Data Scientist-ku wuxuu go'aamiyaa caddaynta loo baahan yahay, hubiyaa in xog ku habboon la hayo, doortaa habka saxda ah, tijaabiyaa natiijada, kadibna sharxaa xaddidaadaha jira.

Mashruuc Kasta oo Data Science ah uma Baahna Machine Learning

Data Science wuxuu isticmaali karaa Machine Learning, laakiin wuxuu sidoo kale adeegsan karaa falanqayn tirakoob, tijaabooyin, ama baaritaan taxaddar leh.



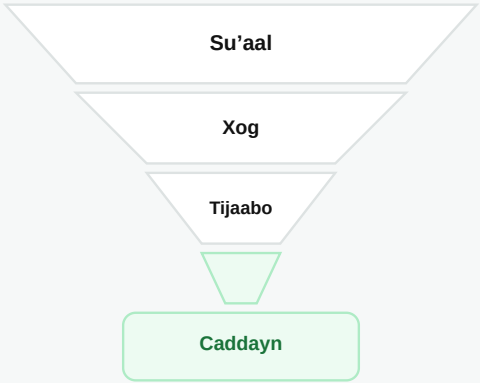
Ka soo qaad in shirkadda isgaarsiintu ay qiimaha xirmooyinka ka beddeshay saddex magaalo xilliyo kala duwan. Data Scientist-ku wuxuu isbarbar dhigi karaa heerka cusboonaysiinta xirmooyinka ka hor iyo ka dib isbeddel kasta, wuxuu baari karaa sababo, wuxuuna hubin karaa in caddayntu muujinayso xiriir. Shaqadaasi waa Data Science xitaa haddii aan la dhisin model Machine Learning ah.

Natiijadu waxay noqon kartaa sharaxaad, tijaabo, saadaal, talo-bixin, ama model. Machine Learning waa hal hab oo la isticmaali karo, laakiin ma aha waxa qeexaya Data Science.

Tusaale

Si loo baaro hoos u dhaca cusboonaysiinta xirmooyinka, Data Scientist-ku wuxuu:

1. Weedha “macaamiishu way baxooyaan” u beddeli karaa su′aal la cabbiri karo
2. Isku dari karaa xogta cusboonaysiinta, qiimaha, iyo adeegga macaamiisha
3. Raadin karaa goorta iyo halka hoos u dhacu ka bilowday
4. Tijaabin karaa waxa sida ugu fiican u sharxaya hoos u dhaca: qiimaha, tayada adeegga, ama arrin kale
5. Soo bandhigi karaa caddaynta, hubanti la′aanta, iyo tallaabooyinka la qaadi karo



Haddii shirkaddu markii dambe doonto inay ogaato macaamiisha gaarka ah ee laga yaabo inay baxaan bisha soo socota, dhibaatadu way is beddeshay. Hadda waxay u baahan tahay saadaal. Halkaas ayaa Machine Learning loogu dari karaa mashruuca.